“Youth Drug Consumption Analysis

24ADI204 - DATA SCIENCE AND VISUALISATION

Submitted by

Lakshmi Prabha M	25BAD049
Anireetha M	25BAD009
Madumitha S	25BAD053
Atchaya S	25BAD016

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

KUMARAGURU COLLEGE OF TECHNOLOGY

(An Autonomous Institution affiliated to Anna University, Chennai)

Post Box No: 2034, Coimbatore - 641049

NOVEMBER 2026

BONAFIDE CERTIFICATE

Certified that this project report “Youth Drug Consumption Analysis ” is the Bonafide work of Atchaya S - 25BAD016, Lakshmi Prabha M M - 25BAD049, Anireetha M - 25BAD009, Madumitha S - 25BAD053 who carried out the project work under my supervision.

SIGNATURE

Dr. Sudharson D

HEAD OF THE DEPARTMENT

Department of Artificial Intelligence and Data Science,

Kumaraguru College of Technology, Coimbatore – 641049.

SIGNATURE

Ms. Shriaarthy E

SUPERVISOR

Department of Artificial Intelligence and Data Science,

Kumaraguru College of Technology, Coimbatore – 641049.

The candidates with college register numbers listed above were examined in the Project Viva-Voce examination held on

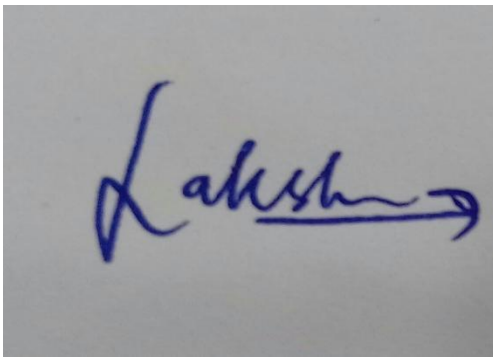
4.11.26

Internal Examiner

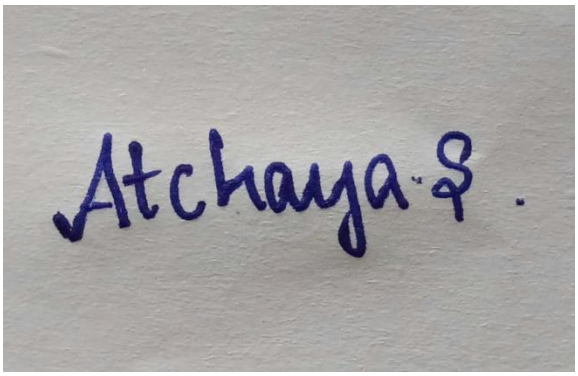
External Examiner

DECLARATION

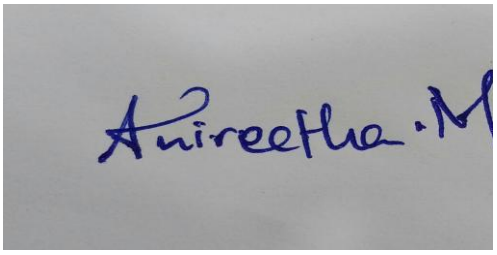
I affirm that the project work titled "**Youth Smoking and Drug Abuse Analysis**" being submitted in partial fulfillment for the award of B. TECH Artificial Intelligence and Data Science is the original work carried out by us. It has not formed the part of any other project work



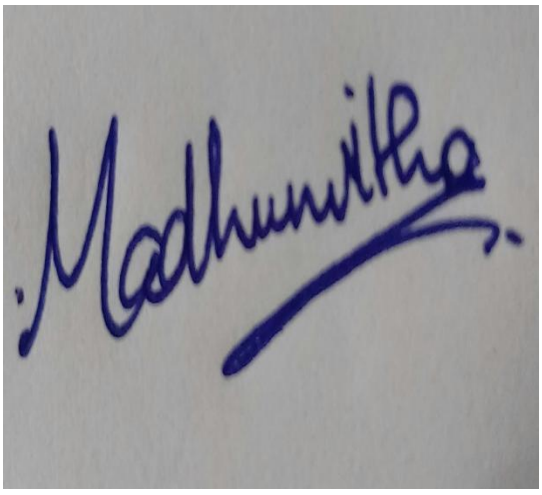
Lakshmi Prabha M M - 25BAD049



Atchaya S - 25BAD016



Anireetha M - 25BAD009



Madumitha S - 25BAD053

Assistant Professor,

Department of Artificial Intelligence and Data Science,

Kumaraguru College of Technology,

Coimbatore – 641049.

TABLE OF CONTENTS

- 1. Executive Summary
- 2. Introduction
- 3. Problem Statement, Objectives and Scope
- 4. Dataset Description and Variable Overview
- 5. Methodology and Project Workflow
- 6. Experiment 1 – Team Formation and Data Hunting
- 7. Experiment 2 – Know Your Data
- 8. Experiment 3 – The Cleaning Sprint
- 9. Experiment 4 – EDA Deep Dive
- 10. Experiment 5 – Feature Engineering
- 11. Experiment 6 – Dimensionality Reduction
- 12. Experiment 7 – Dashboard Construction
- 13. Overall Results and Key Insights
- 14. Challenges and Limitations
- 15. Individual Contributions
- 16. Future Work
- 17. Conclusion
- 18. References

1. EXECUTIVE SUMMARY

This project, titled “Youth Smoking and Drug Consumption: From Raw Data to Visual Insights”, applies a structured data-science workflow to the UCI Drug Consumption dataset. The work covers dataset selection, data inspection, cleaning, exploratory data analysis, feature engineering and dimensionality reduction. An interactive dashboard is the final communication layer intended to bring the findings together.

The original dataset contains 1,885 records and 32 columns. It combines demographic attributes, seven personality-related scores and 19 substance-consumption variables. For this project, participants in the 18–24 age group are treated as the youth segment. This is an analysis definition adopted by the team; the full dataset is not exclusively a youth dataset.

During data cleaning, missing values in Age, Education and Nscore were handled using median imputation, while Gender was handled using mode imputation. The cleaned dataset retained 1,885 records and 32 original columns, with zero missing values and zero duplicate rows. Exploratory analysis used distribution plots, count plots, boxplots, scatter plots and a personality-score correlation heatmap.

Feature engineering included categorical encoding, one-hot encoding, standardization, Min-Max scaling, skewness analysis and binning. Dimensionality reduction was performed on the seven personality features. PCA retained 95.07% cumulative variance using six principal components, reducing the representation from seven features to six.

The project demonstrates how data preparation and visualization can make a complex dataset easier to inspect. The findings are descriptive and should not be interpreted as evidence that personality or demographic attributes cause substance use.

2. INTRODUCTION

Substance-use datasets contain multiple types of information, including demographic attributes, personality measurements and reported consumption categories. Understanding such data requires more than displaying raw values. The data must first be inspected for quality issues, prepared using appropriate techniques and explored through visualizations that match the type of each variable.

This project uses the Drug Consumption dataset from the UCI Machine Learning Repository. The team follows the data-science process from raw-data inspection to data cleaning, exploratory analysis, feature engineering and PCA. The final stage is an interactive dashboard intended to communicate the analysis through a connected visual narrative.

The term “youth” in this report refers specifically to the 18–24 age group selected by the team for focused analysis. Other age groups remain in the dataset and are used for broader comparisons where appropriate.

3. PROBLEM STATEMENT, OBJECTIVES AND SCOPE

3.1 Problem Statement

The selected dataset contains demographic information, personality scores and coded consumption responses for several substances. The variables have different data types and scales, and the original data includes missing values. Without suitable preparation, these characteristics can make comparisons and interpretation difficult. The project therefore aims to inspect, clean, analyze and visualize the dataset in a systematic and transparent manner.

3.2 Objectives

- Understand the structure, attributes and quality of the UCI Drug Consumption dataset.
- Identify missing values, duplicate records, data types and unique-value patterns.
- Apply suitable imputation methods and investigate potential outliers.
- Perform univariate and bivariate exploratory data analysis using appropriate plots.
- Apply encoding, scaling, skewness assessment and binning techniques.
- Use feature-selection checks and PCA to reduce the personality-feature space.
- Develop an interactive dashboard that communicates the analysis clearly.
- Present findings carefully without making unsupported causal claims.

3.3 Scope

The project covers a complete analytical workflow using the selected structured dataset. It includes demographic, personality and substance-consumption variables. The youth-focused analysis uses the 18–24 age group as a project-defined segment. The work is descriptive and exploratory; it does not diagnose individuals, predict an individual's behaviour or establish causal relationships.

4. DATASET DESCRIPTION AND VARIABLE OVERVIEW

The project uses the Drug Consumption dataset available through the UCI Machine Learning Repository. The original dataset has 1,885 observations and 32 columns. It is structured tabular data.

Attribute	Description
Dataset	Drug Consumption dataset
Source	UCI Machine Learning Repository
Original records	1,885
Original columns	32
Data type	Structured / tabular
Variable groups	Demographic, personality and substance-consumption variables
Youth segment used in this project	Age group 18–24

4.1 Demographic Variables

Age, Gender, Education, Country and Ethnicity describe participant characteristics. Some demographic variables are stored as encoded values. Readable labels were created for selected visualizations while retaining the original encoded columns for analysis.

4.2 Personality Variables

The seven numerical personality features are Nscore (Neuroticism), Escore (Extraversion), Oscore (Openness), Ascore (Agreeableness), Cscore (Conscientiousness), Impulsive (Impulsivity) and SS (Sensation Seeking). These variables are used for distribution analysis, scaling, correlation checks and PCA.

4.3 Substance-Consumption Variables

The dataset contains 19 substance-related variables: Alcohol, Amphet, Amyl, Benzos, Caff, Cannabis, Choc, Coke, Crack, Ecstasy, Heroin, Ketamine, Legalh, LSD, Meth, Mushrooms, Nicotine, Semer and VSA. The responses use labels such as CL0–CL6. These are treated as coded response categories. Their precise interpretation should follow the dataset's official codebook; the labels are not assumed to represent equally spaced numerical measurements.

5. METHODOLOGY AND PROJECT WORKFLOW

- Data acquisition: select the UCI Drug Consumption dataset and load it into the Python analysis environment.
- Initial inspection: review rows, columns, shape, data types, descriptive statistics, missing values, duplicates and unique values.
- Data cleaning: create a working copy, impute missing values using data-type-appropriate strategies and investigate potential outliers.
- Exploratory data analysis: study individual distributions and relationships using count plots, histograms, boxplots, scatter plots and correlation analysis.
- Feature engineering: identify numerical and categorical features, encode selected variables, scale numerical features, assess skewness and create bins.
- Dimensionality reduction: apply variance thresholding, correlation analysis, VIF and PCA to the seven personality features.
- Dashboard and communication: integrate the key visual findings into an interactive dashboard and present the results through a report and presentation.
- Python libraries used in the analytical notebook include Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn. Additional statistical utilities were used for outlier and multicollinearity analysis.

6. EXPERIMENT 1 – TEAM FORMATION AND DATA HUNTING

Aim

To form the project team, identify a suitable real-world dataset and establish the initial project workflow and repository.

Procedure

- The team selected the UCI Drug Consumption dataset for its combination of demographic, personality and substance-consumption variables.
- The project focus was defined as exploring consumption-response patterns and presenting the analysis through visual storytelling.
- The dataset source and its variable groups were reviewed before analysis.
- A shared project workflow and GitHub-based organization were planned for storing notebooks, data, reports and presentation materials.

Outcome

The team selected a structured dataset containing 1,885 records and 32 original columns. The dataset was considered suitable for practicing data inspection, cleaning, exploratory visualization, feature engineering and dimensionality reduction.

7. EXPERIMENT 2 – KNOW YOUR DATA

Aim

To inspect the dataset structure, understand its variables and generate an initial data-quality profile before cleaning.

Procedure

- Loaded the dataset using Pandas.
- Displayed the first and last records using head() and tail().
- Checked the dataset shape, column names and data types.
- Generated descriptive statistics for numerical variables.
- Checked missing-value counts, duplicate rows and unique values in each column.
- Reviewed the coded categories used by the substance-consumption variables.

Initial Findings

Inspection	Finding
Dataset shape	1,885 rows × 32 columns
Data types	12 float columns, 1 integer column and 19 object columns
Missing-value columns	Age, Gender, Education and Nscore
Missing values in each listed column	113
Duplicate rows	0
Substance response labels	Coded categories such as CL0–CL6

The inspection showed that the dataset contains a mixture of numerical and categorical attributes. Four columns contained missing values, so data cleaning was required before analysis. The coded substance-response categories were retained as categories rather than interpreted as continuous measurements.

8. EXPERIMENT 3 – THE CLEANING SPRINT

Aim

To improve dataset quality by handling missing values and identifying potential outliers while preserving valid observations.

8.1 Missing-Value Strategy

Variable	Type considered	Method	Reason
Age	Ordinal-coded	Median imputation	Preserves the ordered scale and is less sensitive to extreme values.
Education	Ordinal-coded	Median imputation	Preserves the ordered scale and is less sensitive to extreme values.
Nscore	Numerical score	Median imputation	Provides a robust central-value replacement.
Gender	Categorical / binary	Mode imputation	Uses the most frequent observed category.

A separate working copy of the original dataset was created before imputation. Median imputation was applied to Age, Education and Nscore, while mode imputation was applied to Gender. After imputation, the cleaned dataset contained zero missing values.

8.2 Outlier Detection

Boxplots were used for visual inspection of numerical variables. The Z-score method was also applied, using an absolute Z-score greater than 3 as the criterion for flagging potential outliers. Potential outliers were not automatically deleted because unusual values may represent genuine observations.

Variable	Potential outliers (|Z| > 3)
Nscore	4
Escore	6
Oscore	2
Ascore	5
Cscore	6
Impulsive	7
SS	0

8.3 Data-Quality Result

Check	Result after cleaning
Records	1,885
Original columns	32
Missing values	0
Duplicate rows	0
Potential outliers	Flagged for review; not automatically removed

The cleaning stage improved completeness while preserving the original number of records and columns. Outlier detection was treated as an investigation step rather than an automatic deletion rule.

9. EXPERIMENT 4 – EDA DEEP DIVE

Aim

To explore the distributions of individual variables and examine relationships between selected pairs of variables using suitable visualizations.

9.1 Univariate Analysis

Univariate analysis was used to study one variable at a time. Count plots were used for categorical variables such as Gender, Age Group, Education Level and substance-consumption categories. Histograms were used to examine the distributions of the seven numerical personality scores.

9.2 Bivariate Analysis

Bivariate analysis was used to compare two variables. Grouped count plots were created for Age Group versus Gender, Gender versus Alcohol, Age Group versus Alcohol and Education Level versus Alcohol. Boxplots compared Nscore across Alcohol and Cannabis response categories. Scatter plots examined Nscore against Escore and Oscore. A correlation heatmap summarized pairwise relationships among the personality scores.

9.3 Key EDA Observations

- The dataset contains different variable types, so no single plot type is suitable for every column.
- Count plots are appropriate for comparing frequencies across coded consumption-response categories.
- Histograms and boxplots help inspect the distributions and spread of numerical personality scores.
- Grouped plots allow demographic groups to be compared across response categories.
- The personality correlation heatmap provides an overview of linear relationships among the seven scores.

The EDA stage provided a visual and statistical foundation for the later feature-engineering and dimensionality-reduction stages. The plots describe patterns in this dataset and do not establish causal relationships.

10. EXPERIMENT 5 – FEATURE ENGINEERING

Aim

To prepare the cleaned dataset for further analysis by identifying feature types, encoding selected categorical variables, scaling numerical features, examining skewness and applying binning.

10.1 Feature Identification

Seven personality variables were treated as numerical features: Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS. Nineteen substance-consumption variables were treated as categorical response features.

10.2 Encoding

Label encoding mapped the response labels CL0–CL6 to ordered integer codes 0–6 for the encoding exercise. These codes preserve the category order for computational use; they should not be interpreted as proof that the intervals between categories are equal. One-hot encoding was demonstrated for the Gender_Label variable, producing separate binary indicator columns.

10.3 Scaling

StandardScaler was applied to the seven personality features to transform them to approximately zero mean and unit standard deviation. MinMaxScaler was also applied to rescale these features to a bounded range. The standardization output showed means close to zero and standard deviations close to one.

10.4 Skewness Analysis

Feature	Skewness
Nscore	-0.005123
Escore	-0.001738
Oscore	-0.010308
Ascore	-0.003897
Cscore	-0.005083
Impulsive	0.072795
SS	-0.040660

The absolute skewness values of all seven personality features were below the selected threshold of 0.5. Therefore, no feature was selected for Yeo-Johnson transformation. This is a data-based decision: the transformation was not applied because the chosen skewness criterion was not met.

10.5 Binning

Nscore was discretized using equal-width and equal-frequency binning. Equal-width binning divided the score range into four intervals labelled Low, Medium, High and Very High. Equal-frequency binning divided observations into groups containing approximately similar numbers of records.

10.6 Experiment Summary

Dataset version	Shape
Original cleaned dataset	1,885 × 32
Label-encoded dataset	1,885 × 32
One-hot encoded dataset	1,885 × 33
Standardized dataset	1,885 × 32
Transformed dataset	1,885 × 32
Binned dataset	1,885 × 34

The experiment demonstrated encoding, scaling, skewness assessment and discretization. No Yeo-Johnson transformation was required under the selected threshold.

11. EXPERIMENT 6 – DIMENSIONALITY REDUCTION

Aim

To examine feature relevance and redundancy among the seven personality features and reduce their dimensionality using Principal Component Analysis while retaining most of the variance.

11.1 Feature Selection Checks

- Variance Thresholding was performed with a threshold of 0.01. All seven personality features were retained.
- Correlation-based selection used 0.80 as the threshold for identifying highly correlated feature pairs. No pair crossed this threshold; the highest reported correlation was between Impulsive and SS at approximately 0.623.
- Variance Inflation Factor (VIF) was used to examine multicollinearity. All seven features had VIF values below 5; the maximum reported value was approximately 1.915 for SS.

11.2 Principal Component Analysis

The seven personality features were standardized before PCA because the variables should be placed on comparable scales. PCA was then applied to transform the original features into principal components ordered by the amount of variance they explain.

Component	Explained variance	Cumulative variance
PC1	31.99%	31.99%
PC2	25.77%	57.76%
PC3	12.69%	70.45%
PC4	9.65%	80.11%
PC5	8.16%	88.26%
PC6	6.80%	95.07%
PC7	4.93%	100.00%

Six principal components were required to retain at least 95% of the total variance. The six-component representation retained approximately 95.07% of the variance and reduced the feature representation from seven dimensions to six.

11.3 Experiment Summary

Measure	Result
Input dataset for PCA	1,885 rows × 7 personality features
Features retained after variance thresholding	7
Features removed by variance thresholding	0
PCA components generated	7
Components selected for at least 95% variance	6
Cumulative variance retained	95.07%
Reduced dataset shape	1,885 rows × 6 components

The first two principal components were also visualized in a two-dimensional scatter plot, with gender labels used to distinguish observations. PCA provides a compact representation of the selected personality features; it does not itself explain or predict substance consumption.

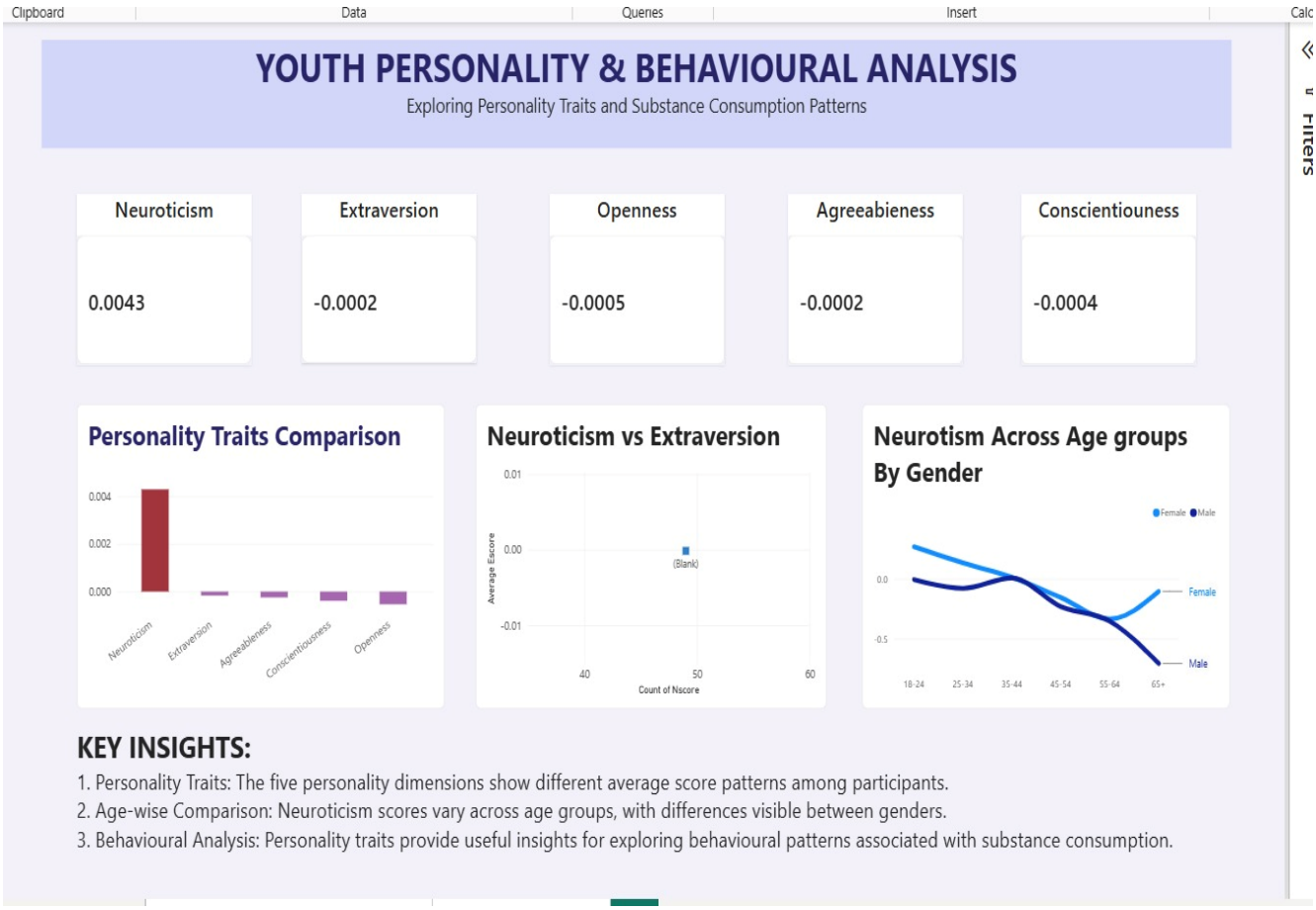
12. EXPERIMENT 7 – DASHBOARD CONSTRUCTION

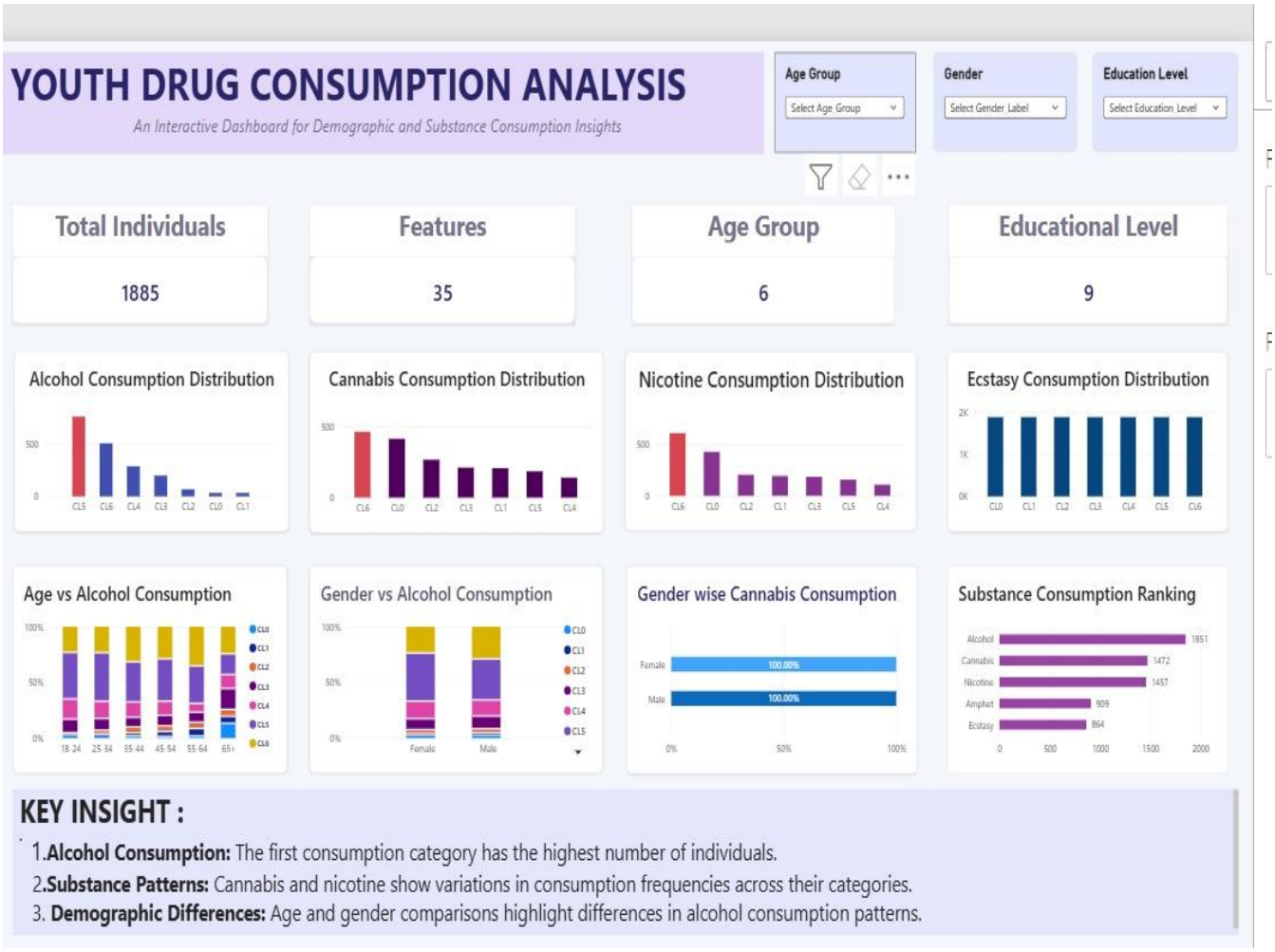
Aim

To integrate the project's key demographic, consumption and personality findings into an interactive visual dashboard that guides viewers from dataset context to analytical observations.

Planned Dashboard Structure

Dashboard page	Proposed content
Project Overview	Participant count, age-group distribution, gender distribution and project-level summary.
Demographic Analysis	Age, gender, education and country distributions with interactive filters.
Consumption Patterns	Distribution of response categories across the 19 substance variables.
Personality Analysis	Personality-score distributions, correlation view and selected scatter plots.
Youth Analysis	Focused analysis of the project-defined 18–24 age group.





The team selected Power BI as the dashboarding platform. The faculty announcement permits Power BI/Tableau as dashboard alternatives to the Streamlit/Plotly approach. The cleaned CSV is used as the dashboard input. The dashboard should include interactive slicers and at least five different visualization types, and its pages should communicate a connected story rather than display unrelated charts.

13. OVERALL RESULTS AND KEY INSIGHTS

- The original dataset contains 1,885 records and 32 columns, covering demographic, personality and 19 substance-consumption variables.
- The cleaning process handled the identified missing values and retained all 1,885 records; the cleaned dataset had zero missing values and zero duplicate rows.
- The 18–24 age group is the project's defined youth segment. In the exported analysis dataset, this group contains 605 participants, approximately 32.1% of the 1,885 records.
- The seven personality variables showed low absolute skewness under the selected 0.5 threshold, so Yeo-Johnson transformation was not required.
- Variance thresholding retained all seven personality features, while correlation and VIF checks did not indicate severe redundancy under the thresholds used.
- PCA reduced the seven personality features to six components while retaining approximately 95.07% cumulative variance.

These results describe the selected dataset and the methods applied by the team. They should not be generalized to all young people or interpreted as evidence of cause and effect.

14. CHALLENGES AND LIMITATIONS

14.1 Challenges Faced

- Understanding the different data types and encoded values in the original dataset.
- Selecting imputation methods suitable for ordinal, numerical and categorical variables.
- Distinguishing potential outliers from genuine observations that should not be deleted automatically.
- Choosing visualizations appropriate for categorical response levels and numerical personality scores.
- Interpreting feature-encoding results without assuming that category codes represent equal numerical distances.
- Explaining PCA results while retaining the distinction between original features and principal components.

14.2 Limitations

- The dataset contains 1,885 records; the results are limited to the available sample and variables.
- The youth segment is defined by the team as ages 18–24; the entire dataset is not a youth-only sample.
- Consumption values are encoded categories. Their meanings should be interpreted using the official dataset documentation.
- The analysis is observational and descriptive; it does not establish causal relationships.
- PCA was applied to the seven personality features, not to every column in the dataset.

15. INDIVIDUAL CONTRIBUTIONS

Complete this table with the actual work performed by each member. The role descriptions below are editable placeholders and must be confirmed by the team before submission.

Team member	Register number	Contribution (confirm/edit)
Lakshmi Prabha M M	25BAD049	Responsible for dataset collection, data inspection, cleaning and data quality validation.
Anireetha M	25BAD009	Responsible for feature engineering, data transformation, correlation analysis and PCA.
Madumitha S	25BAD053	Responsible for data visualization, dashboard development and presentation preparation.
Atchaya S	25BAD016	Responsible for statistical analysis, descriptive statistics and exploratory data analysis.

16. FUTURE WORK

- Complete and validate the Power BI dashboard, including slicers, interactions and final screenshots.
- Use the official UCI documentation to confirm the interpretation of the CL0–CL6 response labels before describing individual categories.
- Improve the dashboard narrative by linking each visualization to a clearly stated question and observation.
- Add a reproducible README file, environment requirements and organized weekly folders to the GitHub repository.
- Extend the analysis only when additional data and suitable validation methods are available.

17. CONCLUSION

The project applied an end-to-end data analysis workflow to the UCI Drug Consumption dataset. The team inspected the original data, identified missing values, applied median and mode imputation, investigated potential outliers and performed univariate and bivariate EDA. Feature engineering introduced encoding, scaling and binning techniques, while skewness analysis guided the decision not to apply Yeo-Johnson transformation. Dimensionality reduction using PCA represented the seven personality features with six components while retaining approximately 95.07% of the variance.

The project demonstrates the importance of understanding data quality and selecting analytical methods according to variable type and purpose. The final dashboard is intended to bring the findings together in an

accessible visual narrative. All conclusions remain descriptive and are limited to the selected dataset.

18. REFERENCES

1. UCI Machine Learning Repository. Drug Consumption (quantified) dataset. https://archive.ics.uci.edu/dataset/373/drug+consumption+quantified
1. Pandas Documentation. https://pandas.pydata.org/docs/
1. NumPy Documentation. https://numpy.org/doc/
1. Matplotlib Documentation. https://matplotlib.org/stable/
1. Seaborn Documentation. https://seaborn.pydata.org/
1. Scikit-learn Documentation. https://scikit-learn.org/stable/
1. Microsoft Learn. Power BI documentation. https://learn.microsoft.com/power-bi/